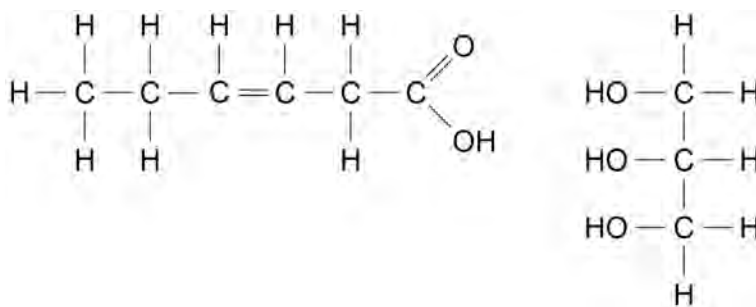


0 1 . 1

Figure 1 shows the structure of a fatty acid molecule and the structure of a glycerol molecule.

Figure 1



On **Figure 1**, **draw a circle** around the part of the fatty acid molecule **and** the part of the glycerol molecule that is removed to form a bond in a triglyceride molecule.

Name the bond formed between a fatty acid and glycerol in a triglyceride molecule.

Name the reaction involved in forming a bond between a fatty acid and glycerol in a triglyceride molecule.

[3 marks]

Bond _____

Reaction _____

Table 1 shows information about three fatty acids.

Table 1

Name of fatty acid	Diagram of fatty acid structure	Fatty acid melting point / °C
Stearic acid	$ \begin{array}{cccccccccccccccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & & & & & & & & & & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} \\ & & & & & & & & & & & & & & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	70
Oleic acid	$ \begin{array}{cccccccccccccccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & & & & & & & & & & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & =\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} \\ & & & & & & & & & & & & & & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	14
Linoleic acid	$ \begin{array}{cccccccccccccccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \\ & & & & & & & & & & & & & & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & =\text{C} & -\text{C} & =\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\text{C} \\ & & & & & & & & & & & & & & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & \text{H} \end{array} $	-5

0 1 . 2 Name the fatty acid shown in **Table 1** that is a saturated fatty acid.

[1 mark]

0 1 . 3 The melting point is the temperature at which a solid changes state to be a liquid.

Use **Table 1** to describe the relationship between fatty acid structure and fatty acid melting point.

[1 mark]

Question 1 continues on the next page

Turn over ►

0 1 . 4

The ratio of saturated to unsaturated fatty acids in a cell-surface membrane determines the extent of the membrane's fluidity.

Scientists provided a cell culture of mouse phagocytes with liquid broth rich in unsaturated fatty acids.

The scientists observed:

- an increase in the proportion of phospholipids in the phagocytes containing unsaturated fatty acids
- more phagocytosis.

Suggest and explain why there was more phagocytosis.

[3 marks]

E

